PURITY ALONG THE FIBRE IS UNIVERSAL FOR DECENT FINITE TYPE MORPHISMS UNDER FLATNESS AND PURITY ABOVE THE POINT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a three step reduction: strictly henselian localization by flat descent, etale local freeness along the fibre, freeness implies universal purity with the relative assassin image criterion closing the loop

Editorial note. The label lemma-finite-type-flat-pure-along-fibre-is-universal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Descent of purity along flat base change: being pure above a point is unaffected by a flat base change of the target
- The etale local freeness along the fibre construction producing an etale neighborhood with an affine cover whose module of sections is free over the strictly henselian local ring
- Affine locally free (projective) modules are universally pure relative to the base
- The two part criterion relating universal purity to the image of the morphism containing the relative assassin

Source reference. Original lemma 05K6: sources/stacks-chow/tags.txt, source lines 7190–7190. Prerequisite pointer invoked at proof line(s) 1137; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05L1: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 781–781. Prerequisite pointer invoked at proof line(s) 1132; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05J9: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 546–546. Prerequisite pointer invoked at proof line(s) 1126; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-flat.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0CW3: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1050–1050. Prerequisite pointer invoked at proof line(s) 1140; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $y \in |Y|$. Assume*

(1) *f is decent and of finite type,*
(2) *$\mathcal{F}$ is of finite type,*
(3) *$\mathcal{F}$ is flat over Y at all points lying over y, and*
(4) *$\mathcal{F}$ is pure above y.*

Then $\mathcal{F}$ is universally pure above y.

Proof. Consider the morphism $\mathrm{Spec}(\mathcal{O}_{Y,\overline{y}}) \to Y$. This is a flat morphism from the spectrum of a strictly henselian local ring which maps the closed point to y. By Lemma 05J9 we reduce to the case described in the next paragraph.

Assume Y is the spectrum of a strictly henselian local ring R with closed point y. By Lemma 05L1 there exists an étale morphism $g : X' \to X$ with $g(|X'|) \supset |X_y|$, with X' affine, and with $\Gamma(X', g^*\mathcal{F})$ a free R-module. Then $g^*\mathcal{F}$ is universally pure relative to Y, see More on Flatness, Lemma 05K6. Hence it suffices to prove that $g(|X'|)$ contains $\mathrm{Ass}_{X/Y}(\mathcal{F})$ by Lemma 0CW3 part (1). This in turn follows from Lemma 0CW3 part (2). □